

Homework 10

1. Find the tangent plane to the unit sphere $x^2 + y^2 + z^2 = 1$ at an arbitrary point (x_0, y_0, z_0) .
2. Let $x, y, f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $f(x, y) = f(x(u, v), y(u, v))$. Prove that

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}.$$

3. Consider the quadratic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = \frac{1}{2}(x^2 + by^2)$ with $b > 0$ and the following gradient descent method for finding its minimum

$$(x_{k+1}, y_{k+1}) = (x_k, y_k) - s_k \nabla f(x_k, y_k),$$

with step size $s_k > 0$ (also called learning rate in the machine learning literature).

- (a) Find the step size s_k using exact line search, such that it minimizes the function

$$\varphi(s_k) = f(x_{k+1}, y_{k+1}) = f((x_k, y_k) - s_k \nabla f(x_k, y_k)).$$

Note that for this optimal step size s_k we have that $\varphi'(s_k) = 0$.

- (b) [Python] For $b = 1, \frac{1}{2}, \frac{1}{5}, \frac{1}{10}$ plot the gradient descent iterations and some relevant contour lines of f to show convergence towards the minimum. What do you notice as b gets smaller?